

Predicting which electricity and gas clients should be inspected first

Fraud detection as a ranking problem

135,493 clients

4,476,749 invoices

15 features

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ML Week 08 Challenge

Today's route

From the business problem to a tested recommendation.

01	Problem and value	Business case, metric choice, and evaluation design	03–06
02	Data and EDA	Aggregation, feature contract, patterns, and bias	07–10
03	Modelling	Pipeline, baseline, extensions, errors, and final model	11–17
04	Recommendation	Operating controls and limits	18
05	Discussion and next steps	Questions, roadmap, and engineering work	19–23

Business case

The useful output is a ranked inspection queue, not a universal yes/no verdict.

Limited inspection capacity

A utility cannot investigate all 135,493 clients.
The model must concentrate genuine fraud near
the top of a finite queue.

Model output

risk score

Stakeholder decision

inspection cut-off

1 • SCORE

One probability per client

2 • RANK

Sort clients by predicted fraud risk

3 • REVIEW

Inspect only the capacity-limited top slice

4 • LEARN

Return investigation outcomes for monitoring and retraining

What does value mean?

Rare-class ranking quality first, operating-point metrics second.

LABELLED CLIENTS

135,493

FRAUD CLIENTS

7,566

FRAUD PREVALENCE

5.58%

ALL-NEGATIVE ACCURACY

94.42%

PRIMARY · PR-AUC

Measures whether rare fraud cases are concentrated near the top across the ranking.

OPERATING POINT · PRECISION@10% / RECALL@10%

Translates the ranking into hit rate and fraud coverage at a fixed inspection capacity.

ROC-AUC remains a secondary ranking diagnostic. Accuracy is not a model-selection metric for this task.

Our data-science question

Can client and invoice history produce a useful ranking — and where does it fail?

1

Feature table

Aggregate invoices to one row per client

2

Out-of-fold comparison

Five-fold stratified predictions for model selection

3

Tune on development

Search only after selecting the model family

4

One final holdout

Freeze the pipeline, then score once

The conclusion first

Use tuned LightGBM to prioritise investigations — with explicit limits.

FINAL HOLDOUT PR-AUC

0.2647

FINAL HOLDOUT ROC-AUC

0.8354

SHORTEST-HISTORY RECALL@10%

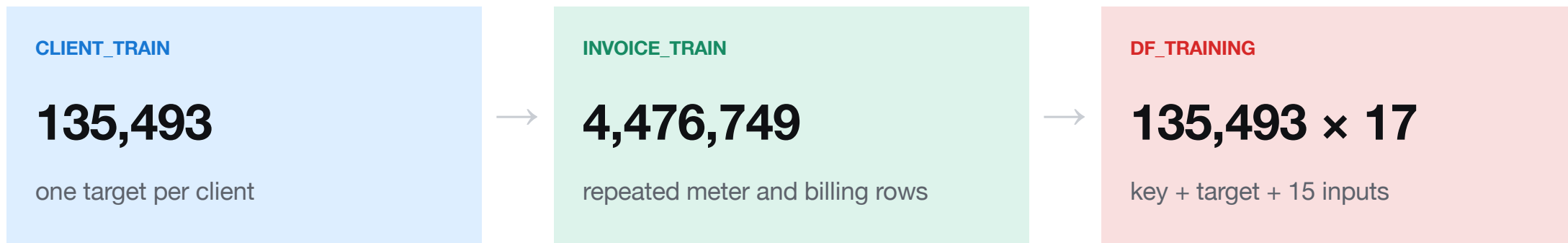
5.6%

- LightGBM materially improves the linear baseline on the same 15-input contract.
- Native categorical handling edges out one-hot encoding in the tuning comparison.
- A LightGBM + Random Forest ensemble is slightly worse and is rejected.
- All three top models miss many of the same fraud clients, suggesting that the current features lack useful information for those cases.

Decision boundary: ranking support + human review + segment monitoring

EDA: from invoices to one prediction row per client

The modelling grain follows the target grain.



MEMORY-AWARE AGGREGATION

Invoice Parquet is processed in 250,000-row chunks. The executed aggregation completed in 16.4 seconds.

OBSERVATION-WINDOW BIAS

Fraud rate rises from 0.94% for clients with 1–7 invoices to 9.15% for clients with 58+. The model may learn how much history exists, not only fraud behaviour.

EDA conclusion: what changes for modelling?

EDA defined the inputs, the evaluation checks, and the next feature work.

MODEL NOW

Use 15 client inputs covering history, consumption, meter sequence, reconciliation, category, district, and region.

CONTROL FOR

Report results by history length and geography. Avoid conclusions based on unreliable creation dates.

INVESTIGATE NEXT

Build stronger short-history signals and validate meter resets and reconciliation gaps with domain experts.

What EDA changed

- The first model uses one explicit 15-input contract rather than every available aggregate.
- Unusual readings remain available as possible signals; rows are not deleted without evidence.

Reproducible modelling pipeline

Deterministic feature creation is separated from learned preprocessing and estimation.



COMMON COMPARISON PIPELINE

Numeric: median imputation + scaling. Categories: one-hot encoding with unknown-category handling.

FINAL LIGHTGBM PIPELINE

Numeric: median imputation + scaling. Categories: native categorical handling; no dummy columns.

REUSABLE FEATURE CREATION

Applies the same client-feature calculations to future raw data. It does not contain the fitted fraud estimator.

Baseline model and results

A simple linear classifier proves the feature contract contains signal.

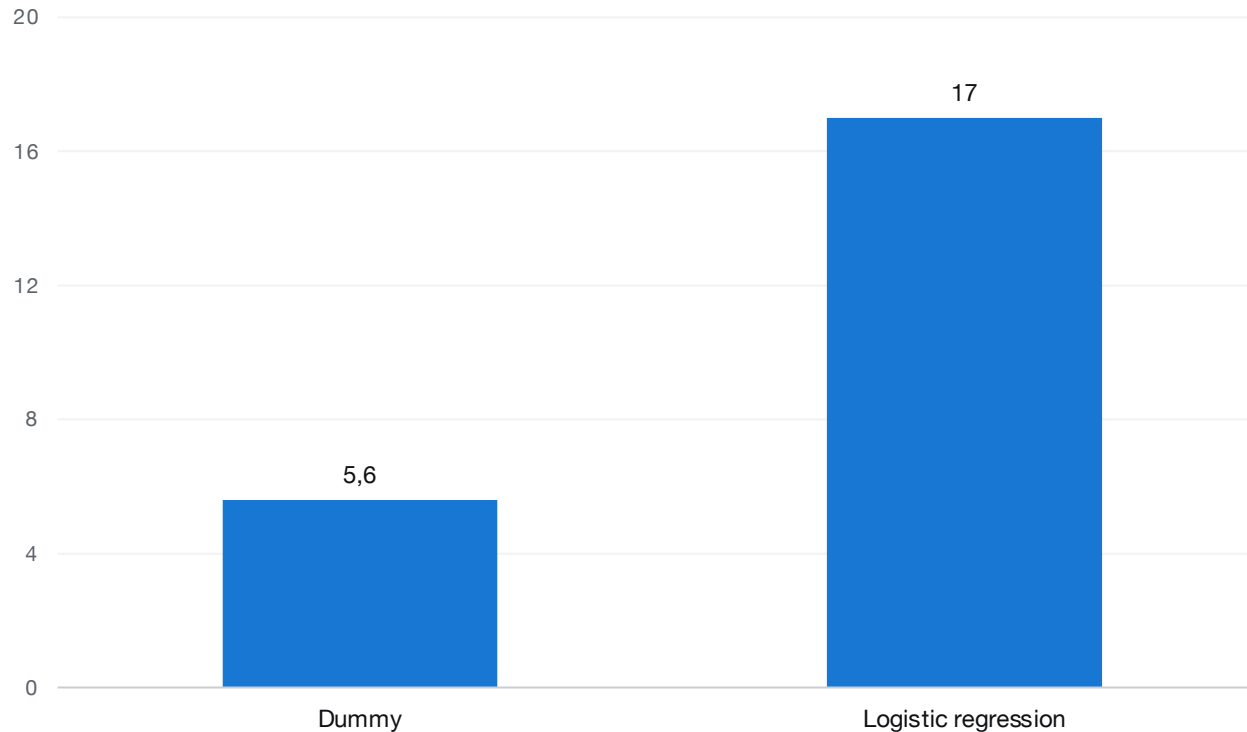


Chart: PR-AUC (%)

DUMMY PR-AUC

0.056

LOGISTIC PR-AUC

0.170

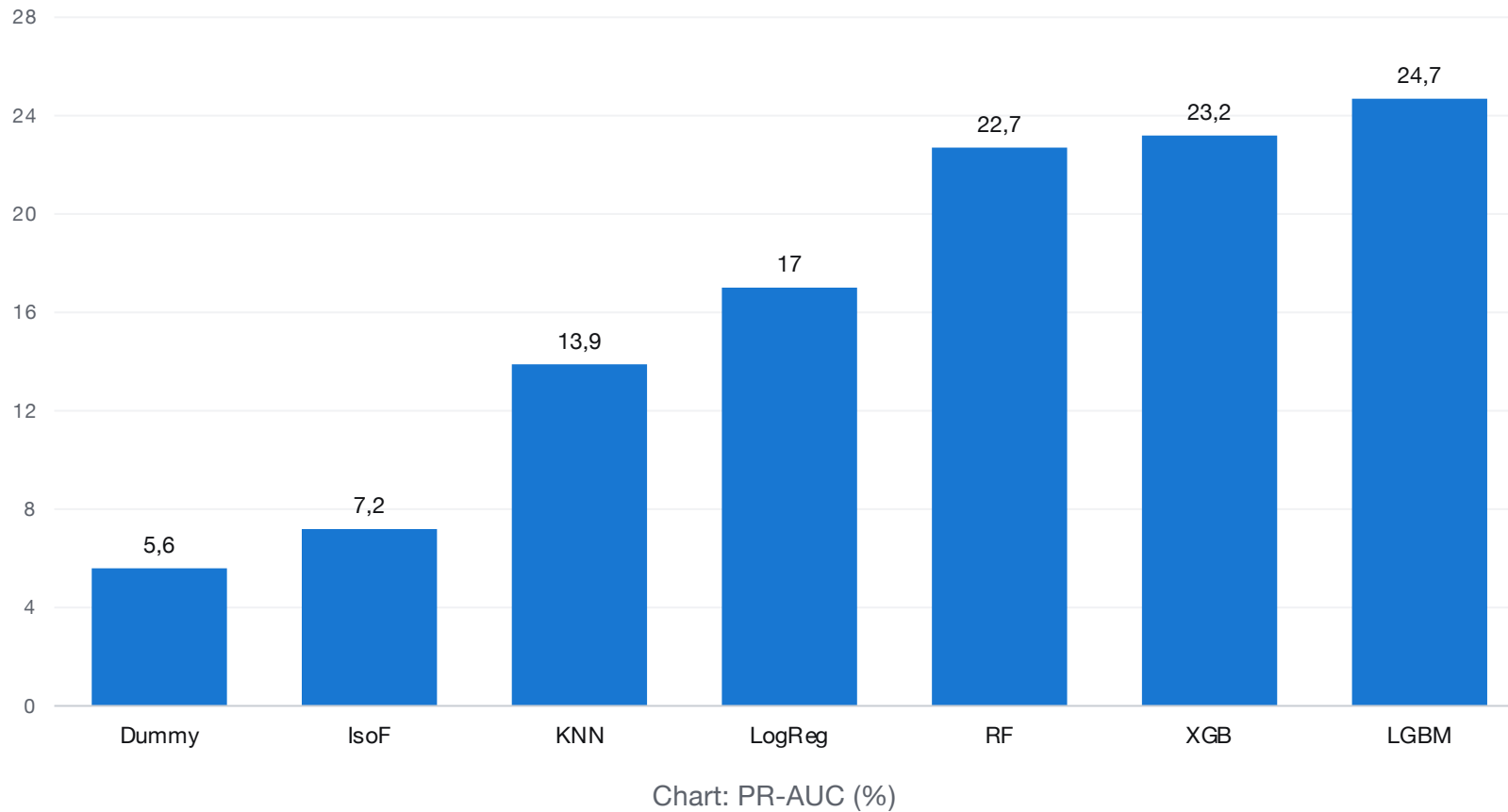
LOGISTIC ROC-AUC

0.795

≈3x PR-AUC lift over prevalence

Model extensions and results

Same inputs and folds, only the model family changes.



WINNER

LightGBM
PR-AUC 0.247
ROC-AUC 0.831

WHY IT WINS

Non-linear splits capture interactions that the linear baseline cannot express.

**Tuning budget goes to
LightGBM**

Error analysis: where the models fail

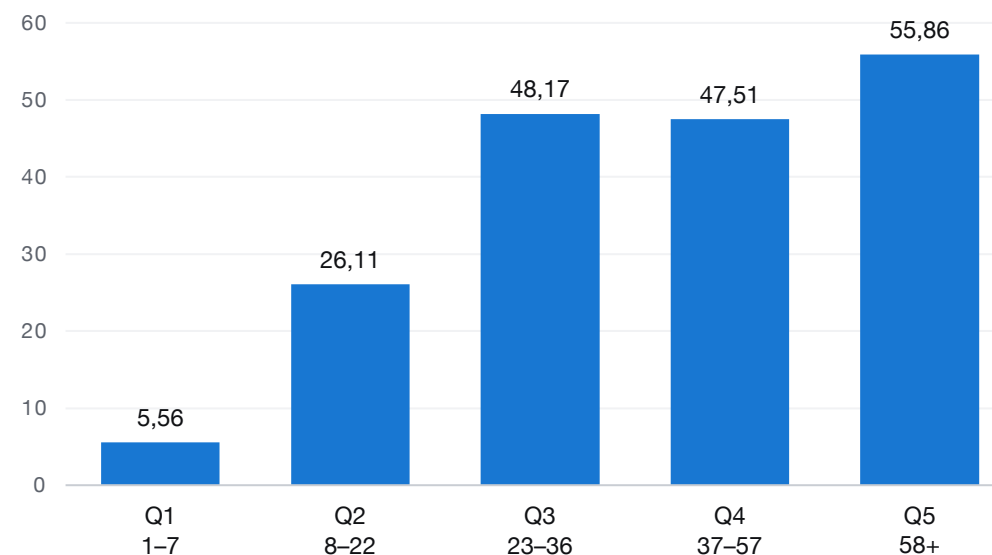
Model agreement exposes a shared blind spot. Holdout segments show where it persists.

Fraud clients by model agreement



3,248 unanimous misses = 42.9% of all fraud

Holdout recall@10% by invoice history



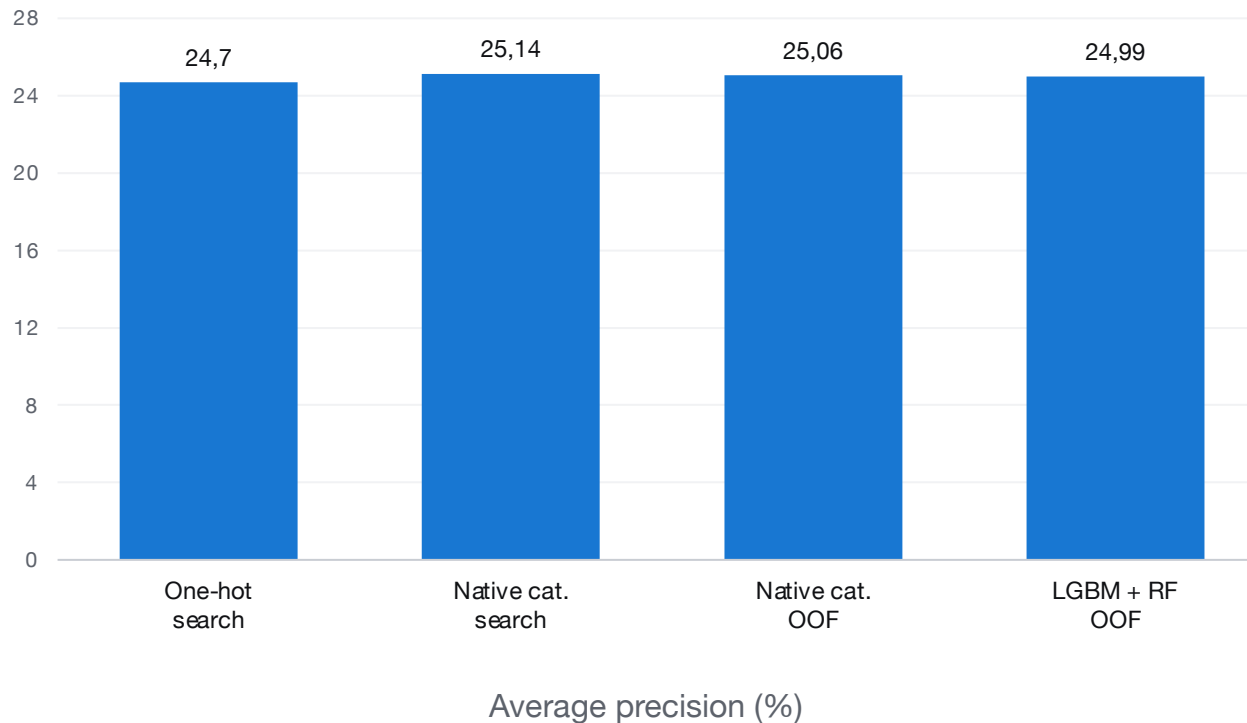
5.6% recall in Q1 → 55.9% in Q5

INTERPRETATION

Shared misses point to missing information in the feature contract.

Final tuned model

Tuning produced a modest gain, evidence rejected the ensemble.



FINAL HOLDOUT PR-AUC

0.2647

FINAL HOLDOUT ROC-AUC

0.8354

MODEL DECISION

Keep tuned LightGBM with native categorical handling. Reject the ensemble: 0.2499 is below 0.2506 and adds complexity.

One illustrative example — not a recommendation

A top-10% threshold makes the trade-off concrete; it is not the chosen production capacity.

TOP 10% OF HOLDOUT

**2,710 of 27,099
clients flagged**

WHAT THAT MEANS

702 fraud caught · 811 missed

2,008 false alarms · 25.9% precision

ACTUALLY FRAUD

702
Caught · 46.4% recall

ACTUALLY NOT FRAUD

2,008
False alarms · 74.1% of
flags

ACTUALLY FRAUD

811
Missed

ACTUALLY NOT FRAUD

23,578
Correctly cleared

ILLUSTRATIVE THRESHOLD ONLY

A real cut-off depends on inspection capacity and the cost of false alarms versus missed fraud.

Zindi submission: a credible first result

The public leaderboard supports the ranking quality without replacing local error analysis.

PUBLIC RANK

275 / 710

PUBLIC SCORE

0.826022

SUBMISSIONS

1

InfoDataChatLeaderboardTeamSubmissions

Submit

Unless stated otherwise in the Info Page, this leaderboard reflects scores based on only a portion of the total test set until the competition closes. See competition Info for more information.

RANK	USER	PUBLIC SCORE	LAST SUBMISSION	# SUBM
275	 sean_kress Go to placement	0.826022007	2 minutes ago	1
1	 +3 Sumy Team	0.901167711	~1 year ago	139
2	 Summerhill-MMAI2024 Team	0.900185082	over 2 years ago	6
3	 stepthom	0.899664465	~4 years ago	15
4	 YusufJimoh_py	0.898408436	~3 years ago	38
5	 Ebiendele Federal university of technology akure	0.898370133	~3 years ago	23
6	 Oracle_	0.897598775	~3 years ago	19
7	 +3 Team Cario Team	0.897398956	~4 years ago	18

Public score $\neq$ local PR-AUC. The leaderboard is provisional and uses Zindi’s competition evaluation.

Recommendation

Adopt the ranking workflow only with operating and monitoring controls.

USE

Tuned LightGBM for inspection prioritisation

CAPACITY

Choose the top-k cut-off from available inspections.

HUMAN REVIEW

Keep an investigator between score and action.

DO NOT USE

Automatic fraud verdicts or causal interpretations

SEGMENTS

Monitor history length, region, district, and category.

FEEDBACK

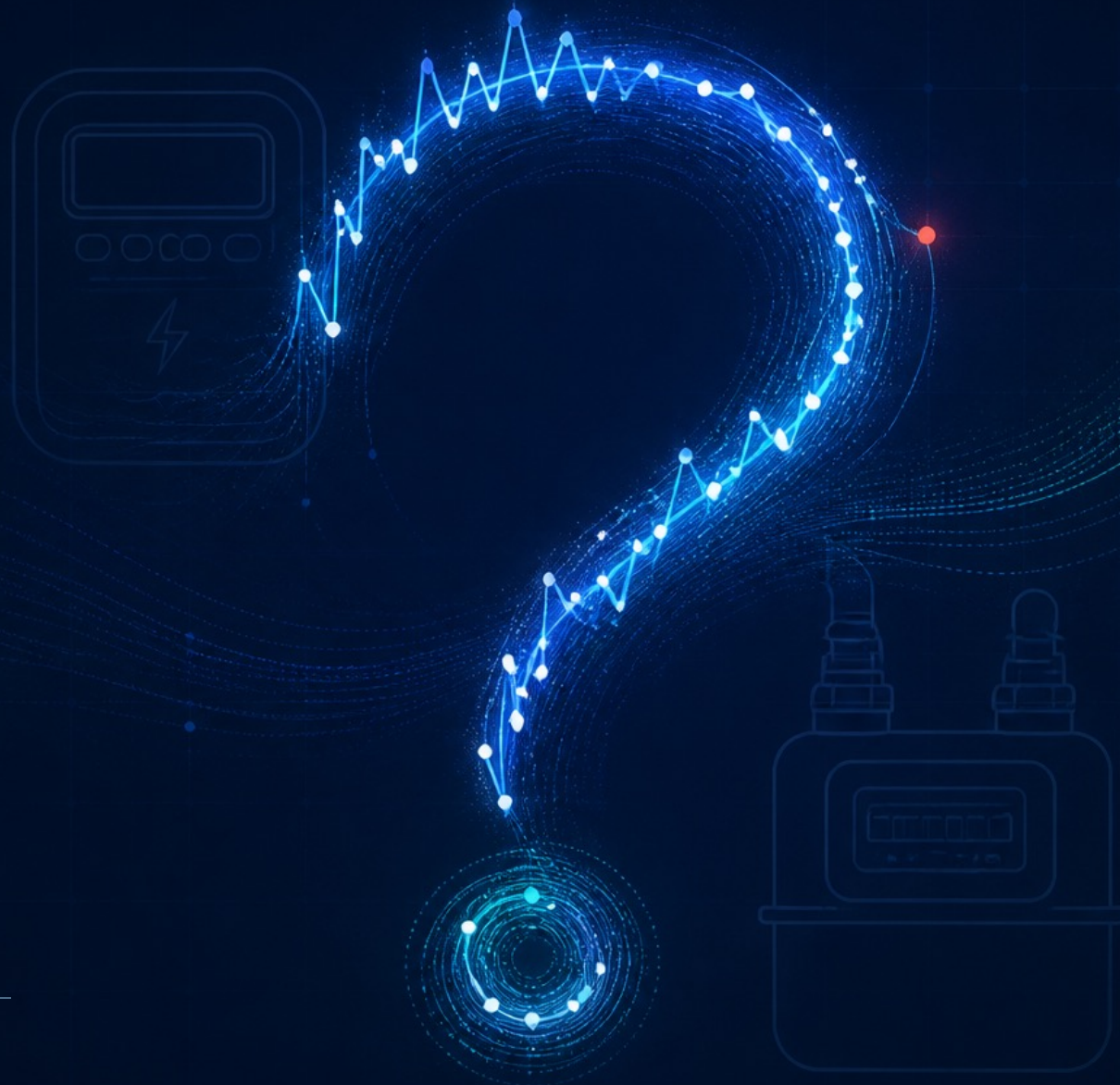
Capture inspection outcomes for calibration and retraining.

ACCEPTANCE CONDITION

Define error costs and minimum segment performance before deployment.

Questions?

Data · Features · Models · Limitations



Extra / bonus: reproducibility and CI/CD

The project is a repeatable workflow, not a single successful notebook run.

01–05

Lifecycle notebooks

EDA → baseline → models → errors → final

PARQUET

Memory-aware data

Headless conversion + chunked aggregation

15

Feature contract

Explicit inputs + reusable transformer

UV

Locked environment

pyproject.toml + uv.lock

DOCKER

Runtime parity

Pinned Jupyter image

CI

Notebook checks

EDA + baseline execute on PRs and main

THANK YOU

Thank you

Roy Grossman · Sean Kress

Fraud detection in electricity and gas consumption



<https://github.com/roy-nfisch/ds-fraud-in-electricity-and-gas>